

A Practical Guide to Independent component analysis

TOPIC -- INDEPENDENT COMPONENT ANALYSIS

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$$\ln L(\mathbf{W}) = \frac{1}{N} \sum_{i=1}^M \sum_{t=1}^N \ln \left(\frac{1}{1 + \tanh(w_i^T x_t)^2} \right) + \ln |\det \mathbf{W}|$$

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since $H(\mathbf{x}) = -\frac{1}{N} \sum_{t=1}^N \ln p(\mathbf{x}_t)$, and maximizing $L(\mathbf{W})$ does not affect $H(\mathbf{x})$, so we can maximize the function

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Linear noiseless ICA The components x_i of the observed random vector $\mathbf{x} = (x_1, \dots, x_m)^T$ are generated as a sum of the independent components s_k , $k = 1, \dots, n$:

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Thus, ICA reduces to finding the orthogonal matrix $\mathbf{V}$. This matrix can be computed using optimization techniques via projection pursuit methods (see Projection Pursuit). Well-known algorithms for ICA include infomax, FastICA, JADE, and kernel-independent component analysis, among others. In general, ICA cannot identify the actual number of source signals, a uniquely correct ordering of the source signals, nor the proper scaling (including sign) of the source signals. ICA is important to blind signal separation and has many practical applications. It is closely related to (or even a special case of) the search for a factorial code of the data, i.e., a new vector-valued representation of each data vector such that it gets uniquely encoded by the resulting code vector

(loss-free coding), but the code components are statistically independent.

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$$K = E \left[\left(\mathbf{y} - \overline{\mathbf{y}} \right)^4 \right] / \left(E \left[\left(\mathbf{y} - \overline{\mathbf{y}} \right)^2 \right] \right)^2 - 3 \quad \{\displaystyle K = \frac{E \left[\left(\mathbf{y} - \overline{\mathbf{y}} \right)^4 \right]}{\left(E \left[\left(\mathbf{y} - \overline{\mathbf{y}} \right)^2 \right] \right)^2} - 3\}$$

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Applications ICA can be extended to analyze non-physical signals. For instance, ICA has been applied to discover discussion topics on a bag of news list archives. Some ICA applications are listed below:

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Rescaling each vector $\mathbf{U}_i = \mathbf{U}_i / \sqrt{E \left[\mathbf{U}_i^2 \right]}$ $\{\displaystyle \mathbf{U}_i = \mathbf{U}_i / \sqrt{E \left[\mathbf{U}_i^2 \right]}\}$, and let $\mathbf{z} = \mathbf{U} \mathbf{z}$ $\{\displaystyle \mathbf{z} = \mathbf{U} \mathbf{z}\}$. The signal extracted by a weighted vector $\mathbf{w}$ $\{\displaystyle \mathbf{w}\}$ is $y = \mathbf{w}^T \mathbf{z}$ $\{\displaystyle y = \mathbf{w}^T \mathbf{z}\}$. If the weight vector $\mathbf{w}$ has unit length, then the variance of y is also 1, that is $E \left[\left(\mathbf{w}^T \mathbf{z} \right)^2 \right] = 1$ $\{\displaystyle E \left[\left(\mathbf{w}^T \mathbf{z} \right)^2 \right] = 1\}$. The kurtosis can thus be written as:

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the kurtosis of the extracted signal y $\{\displaystyle y\}$ to be maximal precisely when $y = s$ $\{\displaystyle y = s\}$. the kurtosis of the extracted signal y $\{\displaystyle y\}$ to be maximal when $\mathbf{w}$ $\{\displaystyle \mathbf{w}\}$ is orthogonal to the projected axes S_1 $\{\displaystyle S_1\}$ or S_2 $\{\displaystyle S_2\}$, because we know the optimal weight vector should be orthogonal to a transformed axis S_1 $\{\displaystyle S_1\}$ or S_2 $\{\displaystyle S_2\}$. For multiple source mixture signals, we can use kurtosis and Gram-Schmidt Orthogonalization (GSO) to recover the signals. Given M signal mixtures in an M -dimensional space, GSO project these data points onto an $(M-1)$ -dimensional space by using the weight vector. We can guarantee the independence of the extracted signals with the use of GSO. In order to find the correct value of $\mathbf{w}$ $\{\displaystyle \mathbf{w}\}$, we can use gradient descent method. We first of all whiten the data, and transform $\mathbf{x}$ $\{\displaystyle \mathbf{x}\}$ into a new mixture $\mathbf{z}$ $\{\displaystyle \mathbf{z}\}$, which has unit variance, and $\mathbf{z} = \left(z_1, z_2, \dots, z_M \right)^T$ $\{\displaystyle \mathbf{z} = (z_1, z_2, \dots, z_M)^T\}$. This process can be achieved by applying Singular value decomposition to $\mathbf{x}$ $\{\displaystyle \mathbf{x}\}$,

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$$h(\mathbf{Y}) = \frac{1}{N} \sum_{t=1}^N \ln \left(\frac{1}{p_{\mathbf{s}}(\mathbf{y}_t)} \right) + \ln |\mathbf{W}| \quad \{\displaystyle h(\mathbf{Y}) = \frac{1}{N} \sum_{t=1}^N \ln p_{\mathbf{s}}(\mathbf{y}_t) + \ln |\mathbf{W}| \}$$

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Minimization of mutual information Maximization of non-Gaussianity The Minimization-of-Mutual information (MMI) family of ICA algorithms uses measures like Kullback-Leibler Divergence and maximum entropy. The non-Gaussianity family of ICA algorithms, motivated by the central limit theorem, uses kurtosis and negentropy. Typical algorithms for ICA use centering (subtract the mean to create a zero mean signal), whitening (usually with the eigenvalue decomposition), and dimensionality reduction as preprocessing steps in order to simplify and reduce the complexity of the problem for the actual iterative algorithm.

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$$p_{\mathbf{Y}}(\mathbf{Y}) = p_{\mathbf{y}}(\mathbf{y}) \left| \frac{\partial \mathbf{Y}}{\partial \mathbf{y}} \right| \quad \{\displaystyle p_{\mathbf{Y}}(\mathbf{Y}) = \frac{p_{\mathbf{y}}(\mathbf{y})}{\left| \frac{\partial \mathbf{Y}}{\partial \mathbf{y}} \right|} \}$$

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In signal processing, independent component analysis (ICA) is a computational method for separating a multivariate signal into additive subcomponents. This is done by assuming that at most one subcomponent is Gaussian and that the subcomponents are statistically independent from each other. ICA was invented by Jeanny Héroult and Christian Jutten in 1985. ICA is a special case of blind source separation. A common example application of ICA is the "cocktail party problem" of listening in on one person's speech in a noisy room.

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The joint pdf $p_{\mathbf{Y}}$ can be shown to be related to the joint pdf $p_{\mathbf{y}}$ of the extracted signals by the multivariate form: